

Actividad 1 — Identificación de la fuente

1. **¿Quién produjo originalmente los datos?** United Nations.
2. **¿Qué organización los procesa para su utilización en Our World in Data?** Our World in Data.
3. **¿Qué representa la variable que analizarás?** El cambio en la población.

Actividad 2 — Carga y exploración de datos

```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: df = pd.read_csv("../data/raw/annual-population-growth/annual-population-growth.csv")
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Entity	Code	Year	Annual change in population	Annual population change (Projected)
0	Afghanistan	AFG	1951	103163.0	NaN
1	Afghanistan	AFG	1952	108441.0	NaN
2	Afghanistan	AFG	1953	108919.0	NaN
3	Afghanistan	AFG	1954	111251.0	NaN
4	Afghanistan	AFG	1955	119027.0	NaN

```
In [4]: df.shape
```

```
Out[4]: (38400, 5)
```

```
In [5]: df.columns.tolist()
```

```
Out[5]: ['Entity',
'Code',
'Year',
'Annual change in population',
'Annual population change (Projected)']
```

```
In [6]: df.dtypes
```

```
Out[6]: Entity          str
        Code           str
        Year           int64
        Annual change in population    float64
        Annual population change (Projected) float64
        dtype: object
```

```
In [7]: df.describe()
```

```
Out[7]:
```

	Year	Annual change in population	Annual population change (Projected)
count	38400.000000	1.868800e+04	1.971200e+04
mean	2025.500000	2.063019e+06	8.375481e+05
std	43.300872	9.440859e+06	5.816745e+06
min	1951.000000	-3.315935e+06	-2.291035e+07
25%	1988.000000	1.781000e+03	-9.742750e+03
50%	2025.500000	4.689800e+04	9.350000e+01
75%	2063.000000	3.021128e+05	1.411160e+05
max	2100.000000	9.337137e+07	7.206736e+07

```
In [8]: df.isnull().sum()
```

```
Out[8]: Entity          0
        Code          1200
        Year          0
        Annual change in population    19712
        Annual population change (Projected) 18688
        dtype: int64
```

```
In [9]: df["Entity"].nunique()
```

```
Out[9]: 256
```

- 1. ¿Cuántas filas y columnas tiene el dataset?** 38,400 filas y 5 columnas.
- 2. ¿Qué tipo de datos contiene la variable Year?** Enteros.
- 3. ¿Existen valores faltantes?** Sí, en las columnas Code (1,200), Annual change in population (19,712) y Annual population change Projected (18,688).
- 4. ¿Cuántas entidades diferentes contiene el dataset?** 256 entidades.

Actividad 3 — Estadística descriptiva

Filtro: **Afghanistan**, período **2005–2023**.

```
In [10]: df_A = df[(df["Year"] >= 2005) & (df["Year"] <= 2023) & (df["Entity"] == "Afghanist")
df_A.head(19)
```

```
Out[10]:
```

	Entity	Code	Year	Annual change in population	Annual population change (Projected)
54	Afghanistan	AFG	2005	843919.0	NaN
55	Afghanistan	AFG	2006	1019525.0	NaN
56	Afghanistan	AFG	2007	485752.0	NaN
57	Afghanistan	AFG	2008	572779.0	NaN
58	Afghanistan	AFG	2009	983471.0	NaN
59	Afghanistan	AFG	2010	817986.0	NaN
60	Afghanistan	AFG	2011	1063620.0	NaN
61	Afghanistan	AFG	2012	1212328.0	NaN
62	Afghanistan	AFG	2013	1062673.0	NaN
63	Afghanistan	AFG	2014	1169819.0	NaN
64	Afghanistan	AFG	2015	1039239.0	NaN
65	Afghanistan	AFG	2016	868846.0	NaN
66	Afghanistan	AFG	2017	988328.0	NaN
67	Afghanistan	AFG	2018	1054099.0	NaN
68	Afghanistan	AFG	2019	1113085.0	NaN
69	Afghanistan	AFG	2020	1212852.0	NaN
70	Afghanistan	AFG	2021	931433.0	NaN
71	Afghanistan	AFG	2022	578437.0	NaN
72	Afghanistan	AFG	2023	875915.0	NaN

```
In [11]: df_A.mean(numeric_only=True)
```

```
Out[11]: Year                2014.000000
Annual change in population    941795.052632
Annual population change (Projected)      NaN
dtype: float64
```

```
In [12]: df_A.median(numeric_only=True)
```

```
Out[12]: Year                2014.0
Annual change in population    988328.0
Annual population change (Projected)      NaN
dtype: float64
```

```
In [13]: df_A.var(numeric_only=True)
```

```
Out[13]: Year                3.166667e+01
Annual change in population  4.446413e+10
Annual population change (Projected)  NaN
dtype: float64
```

```
In [14]: df_A.std(numeric_only=True)
```

```
Out[14]: Year                5.627314
Annual change in population  210865.183261
Annual population change (Projected)  NaN
dtype: float64
```

```
In [15]: df_A.min(numeric_only=True)
```

```
Out[15]: Year                2005.0
Annual change in population  485752.0
Annual population change (Projected)  NaN
dtype: float64
```

```
In [16]: df_A.max(numeric_only=True)
```

```
Out[16]: Year                2023.0
Annual change in population  1212852.0
Annual population change (Projected)  NaN
dtype: float64
```

1. ¿Cuál es el valor de la media? 941,795.05.

2. ¿Cuál es el valor de la mediana? 988,328.

3. ¿Cuál es el valor de la desviación estándar? 210,865.18.

4. ¿Qué entidad presenta el valor mínimo y cuál el máximo? Mínimo: 485,752 (Afghanistan 2007). Máximo: 1,212,852 (Afghanistan 2020).

Actividad 4 — Comprobación del entorno

```
In [17]: import sys
print("Python:", sys.version)
print("Pandas:", pd.__version__)
print("NumPy: ", np.__version__)
```

```
Python: 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD 64)]
Pandas: 3.0.3
NumPy:  2.4.6
```